DOST Form 6

EXECUTIVE SUMMARY FOR ANNUAL PROGRESS REPORT

(Attach DOST Forms 8, 11 and 12)

(1) Program Title: Regional Agri-Aqua Innovation System Enhancement (RAISE) Program in Bicol

Project Title: Project 1-F. Developing the Intellectual Property and Technology Business Management (IP-TBM) in Partido State University through RAISE

Project Leader/Gender: Michael C. Barbecho/Male

Agency: Partido State University.

Address/Telephone/Fax/Email: San Juan Evangelista St., Goa, Camarines Sur

(2) Cooperating Agencies

- Camarines Norte State College (CNSC)
- Camarines Sur Polytechnic Colleges (CSPC)
- Sorsogon State University (SorSU)
- Catanduanes State University (CatSU)
- Central Bicol State University for Agriculture (CBSUA)
- Bicol University (BU)
- Dr. Emilio B. Espinosa, Sr. Memorial State College of Agriculture and Technology (DEBESMSCAT)

(3) Site/s of Implementation (Barangay / Municipality / District / Province / Region / Country)

Base Station: San Juan Evangelista St., Goa, Camarines Sur

Site/s of Implementation:

- Bicol University College of Agriculture and Forestry/ Morera/ Guinobatan/ 2nd District /Albay/ Region V
- ACAD 3 / CoE Room No. 8 Ground Floor Academic Building, F. Pimentel Ave., Barangay II Daet, Camarines Norte
- IP-TBM Office, Center for Research and Development, Nabua, Camarines Sur
- Office of Research, Development, and Extension Services, Magsaysay St., Sorsogon City, Sorsogon
- Catanduanes State University, Calatagan, Virac, Catanduanes 4800
- San Jose, Pili, Camarines Sur
- San Juan Evangelista St., Goa, Camarines Sur
- Cabitan, Mandaon, Masbate

(4) Project Duration (number of months)	**(5) Project Start Date**	**(6) Project End Date**
24 Months	January 1, 2022	December 31, 2023

(7) Major Accomplishments

A. Actual accomplishment of the project (via-a-vis the objectives)

OBJECTIVES	ACCOMPLISHMENTS			
	TARGET	ACTUAL	PERCENTAGE (%)	
			FOR THE PERIOD	CUMULATIVE (FROM START)
General Objectives: To strengthen the capacities of Intellectual Property and Technology Business Management (IP-TBM) of ParSU.	As provided below for each specific objective.	As provided below on a per target basis.	0 to 100%	0 to 100%
Specific Objectives:				
1. Capacitate the technology transfer personnel of ParSU.	2 CMI staff attended Prior Art Search and IP Audit Workshop	Two (2) faculty participated in the Intellectual Property (IP) Master Class webinar last March 2022 to May 2022 with topics on Prior Art Searching: 1. Michael C. Barbecho 2. Jose C. Tunay Jr. III	100%	100%
	2 CMI staff trained in IP MasterClass	Two (2) faculty trained in the Intellectual Property (IP) Master Class: 1. Michael C. Barbecho 2. Jose C. Tunay Jr. III	100%	100%
	2 CMI staff trained in Agribusiness MasterClass	Two (2) faculty trained in the Agribusiness Master Class: 1. Jenny P. Lorio 2. Gina M. Villareal	100%	100%
	2 CMI staff trained in Technology Commercialization Mentorship Series (TCMS)	Two (2) faculty trained in the TCMS: 1. Maria France Q. De Vergara 2. Joel P. Sadol	100%	100%
	2 CMI staff trained in Technology Promotion Mentorship Series (TPMS)	Two (2) faculty trained in the TPMS: 1. Kier P. Gasga 2. Joel P. Sadol	100%	100%
	2 CMI staff attended Communication Plan Workshop	- One (1) faculty attended and participated in the PAHATIRAN: Communication Plan Action Planning last June 14-15, 2022 1. Michael C. Barbecho	150%	150%

		- Three (3) PSU Staff attended the Regional Knowledge Inventory and Communication Planning Workshop on October 24-25, 2022, and December 16, 2022 1. Michael C. Barbecho 2. Melody G. Perias 3. Jella Marie N. Oyales		
	1 Conducted re-echo seminars	Two (2) re-echo seminars conducted on IP, Patent Searching, and Patent Drafting: 1. Seminar on Intellectual Property (IP) Awareness and IP Education (October 27, 2022 - PSU Sagñay Campus | November 4, 2022 – PSU Caramoan Campus | November 7, 2022 - PSU Lagonoy Campus | November 14, 2022 - PSU San Jose Campus) 2. 3-Day Re-echo Seminar Workshop on Patent Searching and Patent Drafting on November 16-18, 2022	200%	200%
	Trained at least 20 CMI staff in re-echo seminars	A total of 81 PSU faculty trained in re-echo seminars conducted on IP Awareness and IP Education, Patent Searching, and Patent Drafting: 1. Seminar on Intellectual Property (IP) Awareness and IP Education (60 participants) 2. 3-Day Re-echo Seminar Workshop on Patent Searching and Patent Drafting (21 participants)	405%	405%
	2 CMI staff attended Policy Webinar/Workshop	Policy Webinar workshop is not available during the Year 1 implementation of the project, but 2 faculty from ParSU had already attended the IP Policy Development Mentorship Series sponsored by DOST PCARRD last	10%	10%

		August 31, 2021 to September 2, 2021		
2. Enhance the technology promotion and commercialization activities of ParSU.	5 Information, Education and Communication (IEC) Materials developed	Two (2) IEC Materials Developed 1. Developing Decorative Items from Abalone Shells 2. Training Module on Intellectual Property (IP) Management, 1st Edition	40%	40%
	5 Intellectual Property (IP) Applications filed	Eight (8) IP applications filed a. Solar/Grid Powered Fish Dryer 1. Invention (1-2022-050436) 2. Industrial Design (3-2022-050799) b. Fish Vendor Cooler 3. Utility Model (2-2022-050727) 4. Industrial Design (3-2022-050797) c. Abalone Shell Christmas Decoration 5. Industrial Design (3-2022-050798) d. Abalone Shell Candle Holder 6. Industrial Design (3-2022-050993) e. Abalone Shell Table Lamp 7. Industrial Design (3-2022-050994) f. Molding Machine for Coconut Leaf (Ibos) Suman Wrapper 8. Industrial Design (3-2022-050992)	160%	160%
	5 Copyright of IECs filed	On-going preparation of required documents for IPOPHL Copyright filing of two (2) IEC materials; For review of the University Knowledge Management Committee	40%	40%

	5 Prior Art Search (PAS) Reports prepared	Five (5) PAS Reports prepared for the following research proposals: 1. Design and Development of Smart Solar/Grid Powered Fish Dryer 2. Designing and Development of Fish Vendor Cooler 3. Development and Fabrication of 3 in 1 Knapsack Agricultural Sprayer 4. In-vitro Mass Propagation in Different Carbon Sources of Taro 5. Development of Metal Foundry Furnace	100%	100%
	1 IP inventory	Conducted one (1) Inventory of potential IP for PSU Tinambac Campus completed researches	100%	100%
	1 Inventory of IP Assets	Conducted one (1) Inventory of IP Assets from year 2014-2022	100%	100%
	1 inventory of matured technologies	Prepared one (1) inventory of matured technologies for Preliminary Formality Exam of DOST V Fairness Opinion Board (FOB) Secretariat	100%	100%
	1 inventory of knowledge resources	Prepared one (1) inventory of knowledge resources, already inputted in the RAISE Bicol Knowledge Inventory Google Sheets	100%	100%
	1 communication plan	Crafted two (1) communication plan with the following title: 1. Local Dried Fish Production in Partido Area 2. Molding Machine for Ibos (Suman) Wrapper	200%	200%
	2 technologies pitched	Two (2) technologies pitched during the Regional Agri-Aqua Technology Pitching Competition on November 29, 2022 1. Coconut Leaf Molding Machine 2. 3-in-1 Rice Cake making Machine	100%	100%

	1 product enhanced (prototyping, developed, packaging, branding)	Identified one (1) technology for prototyping enhancement (3 in 1 Knapsack Agricultural Sprayer)	10%	10%
	1 technology with pre-commercialization reports	Conducted the following preliminary commercialization activities: *12 ParSU Technologies were submitted for Pre-Formality Review by the DOST V Fairness Opinion Board (FOB) last September 28, 2022 *Three (3) proposals were prepared and reviewed by DOST V-FOB for Pre- Commercialization funding under DOST –TAPI TECHNICOM Program 1. Pre- Commercialization of Molding Machine for Coconut Leaf (Ibos) Suman Wrapper 2. Pre-Commercialization of Front Drive Power Train and Suspension Mechanism for Three-Wheeled Electric Vehicle 3. Pre-Commercialization of Hydraulic Leaf Spring Suspension Bushing Remover/Installer	10%	10%
	1 product pre-commercialized manufactured			
	1 Technology Commercialized			
	Participate to the content build-up of RTMS	Consolidated the information of the PSU technologies for commercialization.	100%	100%
3. Intensify linkages with various agencies to enhance activities on intellectual property protection and management and technology transfer and commercialization.	1 Commitment Letter	One (1) commitment letter signed by the agency head and other concerned persons	100%	100%
	1 partnership agreement w/ Business/Trade Institutions	Implementation of Communication Plan for PSU Technologies to target stakeholders to forge agreement with business or trade institutions	0%	0%
	1 Commercialization Agreements	Three (3) proposals were prepared for Pre-Commercialization funding under DOST –TAPI TECHNICOM Program as a preliminary step towards commercialization and signing of agreements	10%	10%

	Crafted/revised IP Policies and Technology Transfer Protocols	- Crafted and developed the PSU IP Policy and was approved by the BOR with Resolution No. 46 s. 2022 (85th Regular Board of Regents Meeting July 21, 2022)	100%	100%
B. Catch-up Plan		**N/A**		
C. Expected Outputs / 6Ps (Expected Outputs should be measurable.)				
	EXPECTED OUTPUTS	**ACTUAL OUTPUTS**		
Publications	5 Information, Education and Communication (IEC) Materials developed	Two (2) IEC Materials Developed 1. Developing Decorative Items from Abalone Shells 2. Training Module on Intellectual Property (IP) Management, 1st Edition		
Patents/IP	5 Intellectual Property (IP) Applications filed	Eight (8) IP applications filed a. Solar/Grid Powered Fish Dryer 1. Invention (1-2022-050436) 2. Industrial Design (3-2022-050799) b. Fish Vendor Cooler 3. Utility Model (2-2022-050727) 4. Industrial Design (3-2022-050797) c. Abalone Shell Christmas Decoration 5. Industrial Design (3-2022-050798) d. Abalone Shell Candle Holder 6. Industrial Design (3-2022-050993) e. Abalone Shell Table Lamp 7. Industrial Design (3-2022-050994) f. Molding Machine for Coconut Leaf (Ibos) Suman Wrapper 8. Industrial Design (3-2022-050992)		
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	1 inventory of knowledge resources	Prepared one (1) inventory of knowledge resources, already inputted in the RAISE Bicol Knowledge Inventory Google Sheets		
	1 communication plan	Crafted two (1) communication plan with the following title: 1. Local Dried Fish Production in Partido Area 2. Molding Machine for Ibos (Suman) Wrapper		

	2 Technologies pitch	Two (2) technologies pitched during the Regional Agri-Aqua Technology Pitching Competition on November 29, 2022 1. Coconut Leaf Molding Machine 2. 3-in-1 Rice Cake making Machine
	1 product enhanced (prototyping, developed, packaging, branding)	Identified one (1) technology for prototyping enhancement (3 in 1 Knapsack Agricultural Sprayer)
	1 technology with pre-commercialization reports 1 product pre-commercialized manufactured 1 Technology Commercialized	Conducted the following preliminary commercialization activities: *12 ParSU Technologies were submitted for Pre-Formality Review by the DOST V Fairness Opinion Board (FOB) last September 28, 2022 *Three (3) proposals were prepared and reviewed by DOST V-FOB for Pre- Commercialization funding under DOST –TAPI TECHNICOM Program 1. Pre- Commercialization of Molding Machine for Coconut Leaf (Ibos) Suman Wrapper 2. Pre-Commercialization of Front Drive Power Train and Suspension Mechanism for Three-Wheeled Electric Vehicle 3. Pre-Commercialization of Hydraulic Leaf Spring Suspension Brushing Remover/Installer
People Services	2 CMI staff attended Prior Art Search and IP Audit Workshop	Two (2) faculty participated in the Intellectual Property (IP) Master Class webinar last March 2022 to May 2022 with topics on Prior Art Searching: 1. Michael C. Barbecho 2. Jose C. Tunay Jr. III
	2 CMI staff trained in IP Master Class	Two (2) faculty trained in the Intellectual Property (IP) Master Class: 1. Michael C. Barbecho 2. Jose C. Tunay Jr. III
	2 CMI staff trained in Agribusiness MasterClass	- Two (2) faculty trained in the Agribusiness Master Class: 1. Jenny P. Lorio 2. Gina M. Villareal
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	Participate to the content build-up of RTMS	Consolidated the information of the PSU technologies for commercialization.
Places and Partnerships	1 Commitment Letter	One (1) commitment letter signed by the agency head and other concerned persons
	1 partnership agreement w/Business/Trade Institutions	Implementation of Communication Plan for PSU Technologies to target stakeholders to forge agreement with business or trade institutions
	1 Commercialization agreement	Three (3) proposals were prepared for Pre-Commercialization funding under DOST –TAPI TECHNICOM Program as a preliminary step towards commercialization and signing of agreements
Policy	Crafted/revised IP Policies and Technology Transfer Protocols	Crafted and developed the PSU IP Policy and was approved by the BOR with Resolution No. 46 s. 2022 (85th Regular Board of Regents Meeting July 21, 2022)

(8) Problems/Concerns

- **Time Constraints of Personnel Involved**
  Personnel involved in the project specifically the project leader/project staff level 3 was academically loaded during the 1st and 2nd quarter of the implementation of the project, affecting the close monitoring of the project.

- **Overloaded Task of Personnel**
  Project Leader/Project Staff Level 3 was overloaded with work due to the absence of Science Research Assistant **(**SRA) resulting to multitasking during the implementation of the project**.**

- **Delayed Fund Download**
  Some of the activities planned such as re-echo trainings, IP filings, and purchase of supplies and equipment were not carried-out on specific schedule due to delayed fund download.

- **COVID-19 Restrictions**
  COVID-19 restrictions prevent the personnel involved in this project from reporting for onsite work to access better internet connection, as a result some of the important meetings and updates regarding the project were missed out.

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(9) Suggested solutions to the above concerns

- **Time Constraints of personnel involved**
  Submit written request for academic de-loading of personnel involve this 3rd and 4th quarter of RAISE IP-TBM project implementation.

- **Overloaded Task of Personnel**
  Request the help and assistance of RAISE IP-TBM Program leader Engr. Christoper Pacardo in hiring qualified Science Research Assistant **(**SRA).

- **Delayed Fund Download**
  Carry out some of the expected outputs and deliverables that does not require funding such as preparation of IP application documents, Prior art searching, IP inventories, IEC material development, etc., while waiting for the fund to be downloaded.

- **COVID-19 Restrictions**
  Look for sites with better internet signal to attend during important meetings and updates regarding the project.

PREPARED BY:

MICHAEL C. BARBECHO
Project Staff Level 3

NOTED:

RAUL G. BRADECINA
President, Partido State University

ANNUAL PROGRESS REPORT

REGIONAL AGRI-AQUA INNOVATION SYSTEM ENHANCEMENT (RAISE) PROGRAM IN BICOL

Project 1-F. Developing the Intellectual Property and Technology Business Management (IP-TBM) in Partido State University through RAISE

MICHAEL C. BARBECHO

Project Leader/Staff Level 3

PARTIDO STATE UNIVERSITY

Implementing Agency

THE PHILIPPINE COUNCIL FOR AGRICULTURE, AQUATIC AND NATURAL RESOURCES RESEARCH AND DEVELOPMENT (DOST- PCAARRD)

Funding Agency

January 1, 2022 – December 31, 2022

Project Title: Project 1-F. Developing the Intellectual Property and Technology Business Management (IP-TBM) in Partido State University (ParSU) through Regional Agri-Aqua Innovation System Enhancement (RAISE)

Project Leader/Project Staff Level 3: Michael C. Barbecho

Implementing Agency/ies: Partido State University-Main

Cooperating Agency/ies:

- Camarines Norte State College (CNSC)
- Camarines Sur Polytechnic Colleges (CSPC)
- Sorsogon State University (SorSU)
- Catanduanes State University (CatSU)
- Central Bicol State University for Agriculture (CBSUA)
- Bicol University (BU)
- Dr. Emilio B. Espinosa, Sr. Memorial State College of Agriculture and Technology (DEBESMSCAT)

Source of Fund: DOST-PCAARD

Budget:	Total approved budget	1,800,000.00 PhP
	Total released budget (as of July 4, 2022)	1,027,549.00 PhP
	Total actual budget expenditures (as of July 31, 2022)	0.00 PhP

Duration:	Actual Start of Project Implementation	(January 1, 2022)
	Actual End of Project Implementation	(December 31, 2023)
	Extension - Start:	(NA)
	Extension - End:	(NA)

Region/Areas Covered: Region V

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LIST OF ACRONYMS/ABBREVIATIONS

AANR - Agriculture Aquatic and Natural Resources
CMI - Consortia Member Institutions
FER - Formality Examination Reports
IPOPHL - Intellectual Property Office of the Philippines
IPMC - Intellectual Property Master Class
IP-TBM - Intellectual Property-Technology Business Management
RDI - Research and Development Institute/Institutions
PAS - Prior Arts Search
ParSU - Partido State University
RAISE - Regional Agri-Aqua Innovation System Enhancement
RTMS - Regional Technology Management System
TCMS - Technology Commercialization Mentorship Series
TPMS - Technology Promotion Mentorship Series

1. INTRODUCTION

The Philippines has further increased its innovation achiever position in the 2020 Global Innovation Index (GII) posting a jump from the 50th place from the 54th in the previous year and is considered the highest position ever of the country. The GII ranks 131 world economies based on innovative capabilities. The innovation inputs and outputs are being considered in the GII.

This achievement has been achieved due to the implementation of various laws that strengthened the scope of protection and support to IP protection and commercialization. One of these laws include the RA 10055: Technology Transfer Act which promote the utilization of intellectual property for the national benefit and shall call upon all research and development institutes and/or institutions (RDIs) that perform government-funded research and development (R&D) to take on technology transfer as their strategic mission and to effectively translate results of government-funded R&D into useful products and services that will redound to the benefit of Filipinos, notwithstanding the income generated from intellectual property rights (IPRs) and technology transfer activities. RA 10372: Revised IP Code further defines and strengthen the IP Code of the Philippines and the Intellectual Property Office of the Philippines to facilitate the protection of products and technologies produced in the Philippines. RA 11293: Innovation Act on the other hand aims to foster innovation as a vital component of national development and sustainable economic growth, while RA 11337: Innovative Start-up Act is to foster inclusive growth through an innovative economy by streamlining government and nongovernment initiatives, in both local and international spheres, to create new jobs and opportunities, improve production, and advance innovation and trade in the country.

DOST as one of the identified “start-up enablers” in the country has made significant contribution to this achievement though various initiatives. In 2016, PCAARRD launched the DOST-PCAARRD Innovation and Technology Center (DPITC) that caters to the technology commercialization initiatives of the agency. Under the initiatives of the DPITC, the Intellectual Property-Technology Business Management (IP-TBM) program was conceptualized and approved. For the batch 1 of the program 5 licensing agreements were executed against the target of 10. for Batch 2, exploratory meetings with potential investors are still on-going. To sustain and pursue the initial efforts in protecting the technologies generated by the participating HEIs in the Batch 1 and 2 of the IP-TBM program. The SUSTAIN IP-TBM Phase II was implemented using the mentor-mentee approach where there are five mentor agencies that taught and guided thirty mentor agencies on IP Protection and Commercialization.

In Bicol region, Patido State University (ParSU) integrated the Research and Development function since 1994. This integral component to be functional must conduct researches, projects and programs responsive and directed towards enhancement of the existing programs of the university and supportive to the country's development goals. Research management at ParSU must enunciate clearly its vision, mission goals and objectives, areas of concern, thrusts and priorities, organizational structure and functions, and implementing mechanisms. This is done to ensure an orderly manner and professional conduct of research in this academic community.

ParSU evolved and implemented an effective mechanism for forecasting, planning, policy making, financing managing, and assessing Research Activities designed to promote the institutions growth and development along the four goals of higher education: access and equity, quality and excellence, relevance and responsiveness, and efficiency and effectiveness, and along the priority areas of the local government and national government. It also want to gain recognition as a reputable

Research unit with a track record of conducting Research activities that will contribute to the pool of knowledge, and solve the problem of the academe, community

ParsU is also conducting research and development studies in social sciences, business, science and technology, culture and arts, agriculture and fisheries. It will strengthen the manpower and resource capabilities of the university. It will formulate and recommend research policies and guidelines conducive to and supportive of the research endeavor of the faculty and other members of the academe. It will establish linkage with government and non-government agencies, industry and business sectors, sister SUC's and other local and foreign institution for fund sourcing, capability building, and networking along the research and development endeavor. ParSU will also disseminate information and technologies responsive to the emerging needs of its clientele. It will also create mechanisms for monitoring and evaluating researches and research activities, and for dissemination, implementation and utilization of research outputs.

With the aforementioned vision, ParSU sees the need seminars and trainings on IP protection and Commercialization that will address the lacking avenues for technology partnerships and commercialization of AANR products of the institution. The RAISE Program can provide the capability training and building on technology transfer to R&D partners while developing ParSU's establishment of technology transfer office. This will also bring forth a maximized economic potential of science and technology innovations.

With the aforementioned vision, ParSU sees the need seminars and trainings on IP protection and Commercialization that will address the lacking avenues for technology partnerships and commercialization of AANR products of the institution. The RAISE Program can provide the capability training and building on technology transfer to R&D partners while developing ParSU's establishment of technology transfer office. This will also bring forth a maximized economic potential of science and technology innovations.

General Objectives:

To strengthen the capacities of intellectual property and technology business management (IP-TBM) operations of PARSU.

Specific Objectives:

1. Capacitate the technology transfer personnel of PARSU;
2. Enhance the technology promotion and commercialization activities of PARSU; and
3. Intensify linkages with various agencies to enhance activities on intellectual property protection and management and technology transfer and commercialization.

2. METHODOLOGY

A. Regional Capacity Building and Mentorship

Training and Capacity Building on IP Protection and Commercialization

The RAISE program has four (4) project wherein Project 1 is the Regionalization of Intellectual Property – Technology Business Management (IP-TBM), Project 2 is the Regional Agribusiness Hub, Project 3 is the Regional Agri-Aqua Technology Business Incubation, Project 4 is the Regional Knowledge Management. These four (4) projects will be based in mentor agency.

The Partido State University will select faculty and students that will be capacitated on IP management and business development/technology commercialization through the conduct of the condensed 10-module training workshop (Figure 1). The university will sent faculty and students to be mentored in technology transfer which will consist of 2 sets of training – 1) the Intellectual Property Master Class, and 2) the Technology Commercialization Mentorship Series. Both sets of training will consist of five (5) modules that will be conducted for 15 days spread over 6 months (3 days/month). The IP Master Class will demonstrate the whole patent application flow through hands-on prior art searches, actual claim drafting, filing application, substantive examination request, responding to Formality Examination Reports (FER) up to the administration of a mock exam for patent agent aspirants. The Technology Commercialization Mentorship Series will equip the trainees with the different pre-commercialization business processes, technology pitching, negotiation, and commercialization strategies. The trainees are expected to file patent or utility model applications during the training, while the prosecution of these applications will be done through mentorship by the trainers.

A training provider that will match the expertise needed will be approached to handle the IP Management training through a government-to-government contract of service. If this is not workable, public bidding will be conducted. For the business development/technology commercialization training, qualified training providers/a pool of experts will be tapped to perform the mentorship series. Contracting will be through public bidding. The series of trainings will be conducted at the identified venue in Bicol.

The project 1 team of RAISE program are composed of the personnel exposed to IP/Technology Commercialization training/conferences in coordination with PCAARRD and IPOPHL. The project will look for possible training providers where IP management and technology commercialization activities are well entrenched and practiced. Below is the training module for the IP protection and Commercialization Mentorship Series.

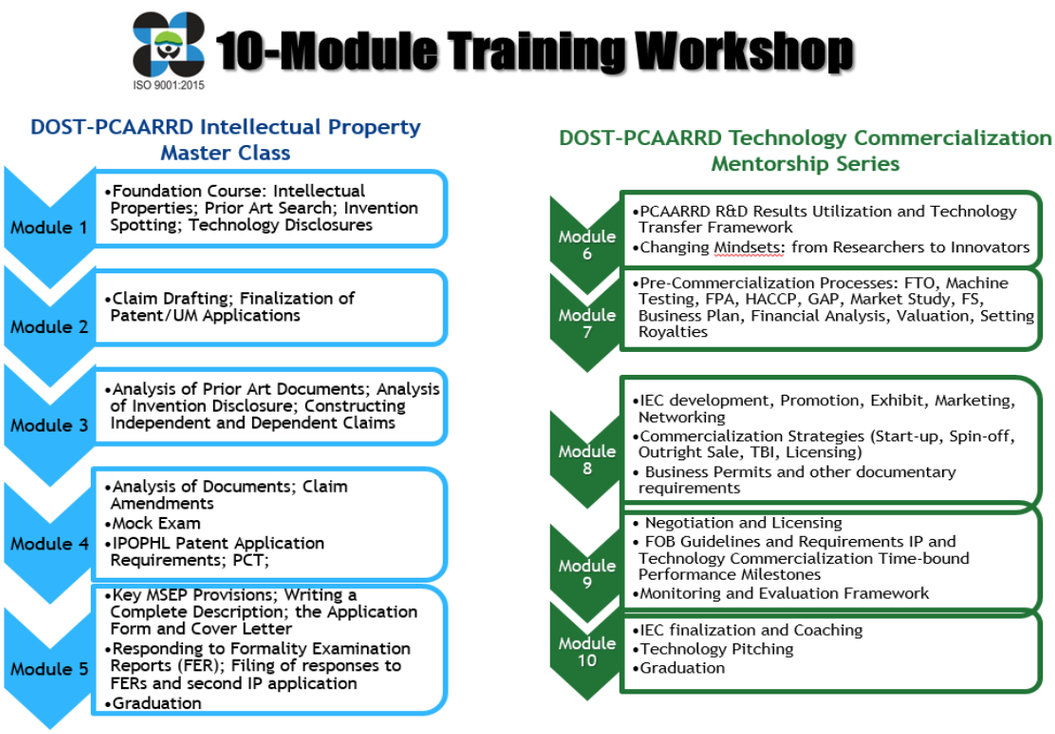


Figure 1. The 10-module Training Workshop on Intellectual Property Management and Technology Business Management that will be adopted by RAISE program.

Partido State University will also conduct the same mentorship to its faculty and students as the participants of the mentorship series will serve as its resource speakers.

To measure the effectiveness of the training/mentorship workshops, each participating institution will be assessed according to the following indicators, but not limited to:

- Number of IP filings
- Number of echo seminars conducted as a resource person
- Number of trainees graduated

The program team shall develop a merit system among the graduates taking into consideration the commitment of the participants for the continued and real practice as patent agents.

Training and Capacity Building on Agribusiness Ventures

Partido State University will participate in the training and capacity building in the creation of the Agribusiness Start-up Companies. This will capacitate faculty participants in the establishment and operation of an agribusiness. The training will be done simultaneously with the IP master Class wherein the target participants are those mentee and mentor who have completed the IP Master Class and IP commercialization mentorship Series. Below is the proposed module of the Training and Capacity Building for the Establishment and Operation of Start-up Company.

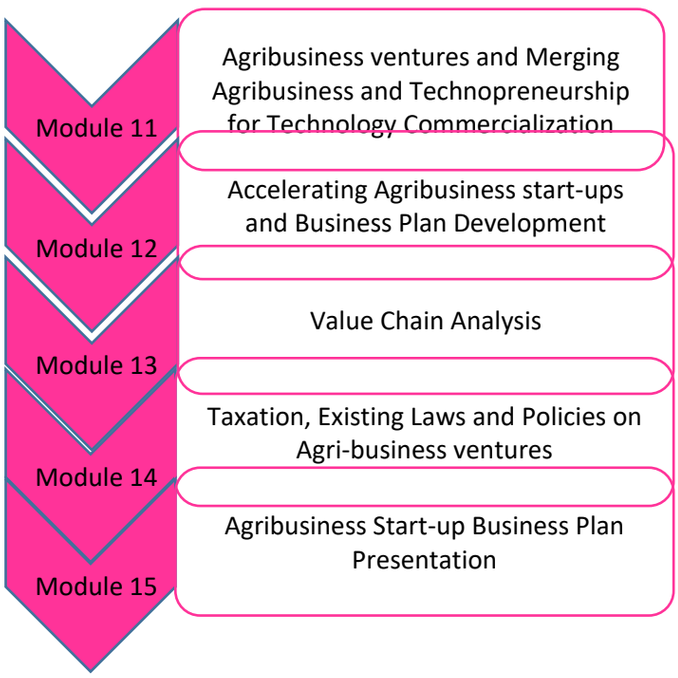


Figure 2. The propose 5-module Training Workshop on Agribusiness Ventures

B. Agri Technology Business Development

Identify New Agricultural Technology that has a Viable Product for Mass Production and Marketing.

Partido State University will identify of at least two (2) agricultural technologies that has a viable product that can be commercialized and marketed in both local, national and international market. The inventors or makers of the technology will be mentored in IP protection

and commercialization the technology through the 10-module Training Workshop on Intellectual Property Management and Technology Business Management of PCAARRD that will be conducted by the mentee agency. The inventors will be partnered with a graduating BS Entrepreneurship Student to help them conduct the feasibility study and make a business model for the product.

Establishment of Start-up Company from Agri-Aqua Technologies

The RAISE program will assist and train technology developers of Partido State University on valuing their developed technology through a series of trainings on IP valuation that will include the knowledge on valuation, conducting market study and generating market data. They will also assist and train technology developers to identify their users, analyzing the costumer journey, costumers empathy mapping, naming the company and organizing their business team, creating a company logo and tag line, creating a company vision, mission and goal, securing license to operate, analyzing and creating value proposition, technology valuation analysis and computation, technology competitive analysis, business model and the use of business model canvass, go-to-market plan and market share analysis, creating business plan as well as assist and train technology developer in analyzing value chain of their raw materials and product.

The RAISE program also will assist and train technology developers on crafting term sheets and licensing agreements for technology commercialization. Assist them in Technology Readiness Level Assessment, getting the Fairness Opinion Report and Attract start-up companies offering agri-aqua technologies to undergo incubation.

Conduct of Product Display, Advertisement and Marketing.

The RAISE program will identify a building and site in Partido State University that will be the display center of the technologies developed by the mentor and mentee agency. It will also urge agri-aqua technology inventors and makers to produce their product and display it to the Agribusiness hub located in the mentor and mentee agencies. It will also urge the agri-aqua inventors and makers to develop an advertising video to be uploaded to the social medias and to be played in the legitimate media outfits. It will further urge preliminary marketing of the product for market testing.

The Incubation of the Selected Technologies

The RAISE project will focus on maximizing and advancing the potential technologies and innovations generated by the faculty, researchers and students of mentor and mentee as well as Partido State University through technology transfer and commercialization by spin-off, licensing, or start-ups. The program will tap and form a pool of technology commercialization advocates, industry collaborators, seed funding partners, and technology and business experts to be the lead promoters in diffusing technology and innovation to the market.

Technology Shopping

The Technology Portfolios will be selected from the IP asset inventory will be identified for incubation. These technologies are from agri-aqua production, agri-aqua equipment/device, agri-aqua food and feed meal, and postharvest technologies.

Business Modelling

A business plan will be developed to the identified technologies for incubation. The Business Model Canvas will be used a device to make sure that the technology or product will surely reach market and the end user and to make it sustainable. The Business Team of the incubatees will be formed and trained on product and service management, promotions, and knowledge management.

Application

Sorsogon State Univesity will endorsed a start-up that will become an incubatee of the Regional ATBI located in the mentor agency. The prospect incubatee will submit an expression of interest in a form of letter will be submitted to the Partido State University and endorsed to Bicol Agri-Aqua Technology Business Incubation office. It will be endorsement to the mentor agency, Dean, Director of Technology Commercialization, Vice President for Research, Development and Extension and to the University President for Approval. After the approval from the President the prospective incubatee will be required to submit their *Business Model Canvas* and *Team Formation. A Memorandum of Agreement will be signed together with the* Non-Disclosure Agreement. This incubatee will become a co-incubatee of the Partido State University and the mentor agency once the university is capable of handling the project.

Incubation

The Partido State University will be endorsed at least two start-up companies to become an experimental incubatee for ParSU and an outwall co-incubatee of the Bicol Agri-Aqua Technology Business Incubation (BATBI). The RAISE program and BATBI will assist in finding partners and collaborators until such time that will be able to execute a commercialization agreement with a business entity or an institutional agreement with an HEI or an RDI. The incubatees of the ParSU and BATBI will be thought with Founders Basic Training, Skill Sets Training, Business Plan Creation and Product/Service Development. Business Mentoring will also be done from time to time. Technology expert assistance, office space and workstations will be provided to incubatee. Networking and Industry Matching will also be done from time to time. Fund Sourcing will also be explored. Partido State University will be learning the process and activities of handling ATBI so that it can establish its own ATBI in the coming years.

Graduation

An incubatee will be classified as graduate they had reach the first million income. A Sales Report, Client Portfolio and Business/Asset Valuation will be required to be submitted every year to know if an incubatee will be classified as graduate.

Post-Graduation

The support to the incubatee who graduated from incubation will not end after graduation. The BATBI will still monitoring, give training and business support networking to the graduated incubatee for them to become more sustaining company.

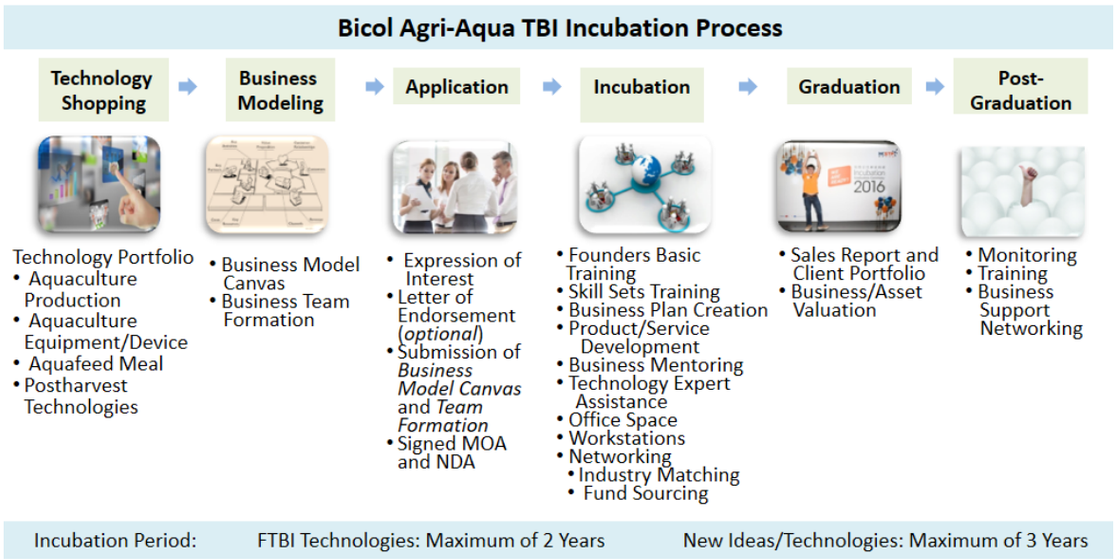


Figure 3. The Bicol Agri-Aqua TBI incubation process

C. Intellectual Property Management

The RAISE program will be dealing with IP Management that will facilitate the process of technology protection. Every year the project will conduct IP Audit to identify and list the technologies that need to be protected. The technologies identified for protection will be subjected to patent search, patent drafting, patent application and prosecution. Ten (10) protected IP's will be the target of the mentor and mentee agencies for 2 years The IP's applied for protection will subjected for commercialization.

IP Disclosure

Researchers and inventors of the CMI's will be requested to submit disclosure of the agri-aqua technologies to the IP-TBM Team of the ParSU especially those technologies generated by through research funded by DOST and DOST PCAARRD. The disclosures will be the basis for assessing the output technology with its novelty, obviousness, and industrial application.

Patent Search

The disclosures will be subjected to patent search. The RAISE program will subscribe to a patent search database to make the patent search report more reliable. The patent search report will be used as basis for drafting the patent of the technology disclosed.

Patent Drafting

The patent will be drafted by the IP TBM team of the Sosogon State University. It will be checked by the Project leader before submitting to IPOPhil. The project leader will endorse its application for IP protection.

IP Application

The RAISE program will fund the IP application of agri-aqua technologies in IPOPHL. The IP TBM staff will be the in-charge of its prosecution.

Commercialization of IP's Generated by Project Funded by PCAARRD

The Intellectual Property Commercialization will be done by establishing a start-up and incubation through the assistance of the mentor agency. Inventors will be partnered with graduating students from agribusiness or entrepreneurship to put up a company top commercialized the generated IP.

IP valuation and Socio-Economic Analysis

The project 2 will tap the SERDAC project to come up with IP valuation and Socio-Economic analysis and computation of the IP to be commercialized.

D. Strategic Partnership and Collaboration

Manage Business Network/Partnerships

Exploratory meetings with potential business partners, such as but not limited to business groups, marketing, and trader's groups will be conducted. Discussions will be made with the local Chamber of Commerce, Inc. (CCI) organization to booster the commercialization of the technologies developed by mentor and mentee agency.

The RAISE program will be targeting the private sector as potential takers of the technologies that they will promote, this partnership will be fully explored to fast-track commercialization of PCAARRD-supported technologies in the SUCs/RDIs through the IP-TBM. Linkages and partnership with the industries, private sector, and other stakeholders. Collaboration with the private sectors, industries local government units, and associations will be made as partners and or prospective adaptors of the commerciable technologies generated by the participating agencies. Inter-personal visits and dialogue with the target partners and potential adopters will be conducted.

To promote the technologies developed by the SUCs/RDIs, networking cum product promotion activities will be conducted or participated by each SUC/RDI. Attendance to the NSTW, PCAARRD SiPAG FIESTA, DOST Technology Transfer Day, etc. will also be encouraged. Potential technology adopters, as well as other collaborators and stakeholders, will be invited to attend the said activity. It is expected that the technologies developed would be able to reach the target adopters. Investment primers and flyers/leaflets of the commerciable technologies will be made to serve as promotional materials.

Also, in the second year, each RAISE will conduct an echo seminar/training to faculty researchers to further disseminate the learnings generated from this project. The IP-TBM staff will serve as resource persons to these echo-seminars.

E. Enhance Knowledge Management

IP Audit

The IP audit will be conducted by each participating agency to determine the types and number of IPs generated that can be included in IP protection and commercialization activities. Each participating agency is expected to submit an Inventory of IP assets and will be updated yearly to be assisted by the mentee agency.

Patent Mining

Patent mining will be done by the mentee agency and ParSU IP TBM staff will assist researchers know the technologies that should be develop concerning the selected commodity. The program will endorse AANR technologies for development as priority agenda for R&D. It will also conduct training on patent mining that the participants are the selected faculty and personnel of the mentor and mentee agencies.

Research Publication and Presentation to R&D For a Monitoring

Research to be published or for presentation to various fora will be requested to pass through RAISE office to be checked by IP TBM staff if there are possible significant disclosure of generated IP. This will be done to avoid public disclosure of IP's before its registration.

Research Inventory

The RAISE program will also be conducting an AANR research inventory every year. This Research inventory will be submitted to BCAARRD and PCAARRD by the ParSU IP TBM staff and will be encoded into the database of the RTMS.

Regional Technology Management System

A regional technology management system (RTMS) will be develop and will serve as a repository of different technologies from the mentor and mentee agencies. This website will be used by the faculty, students, and researchers as a source of related literature and studies and in patent searching. Though this RTMS there will be no duplication and re-inventing the wheel among the researches and technologies in the region and in the entire nation.

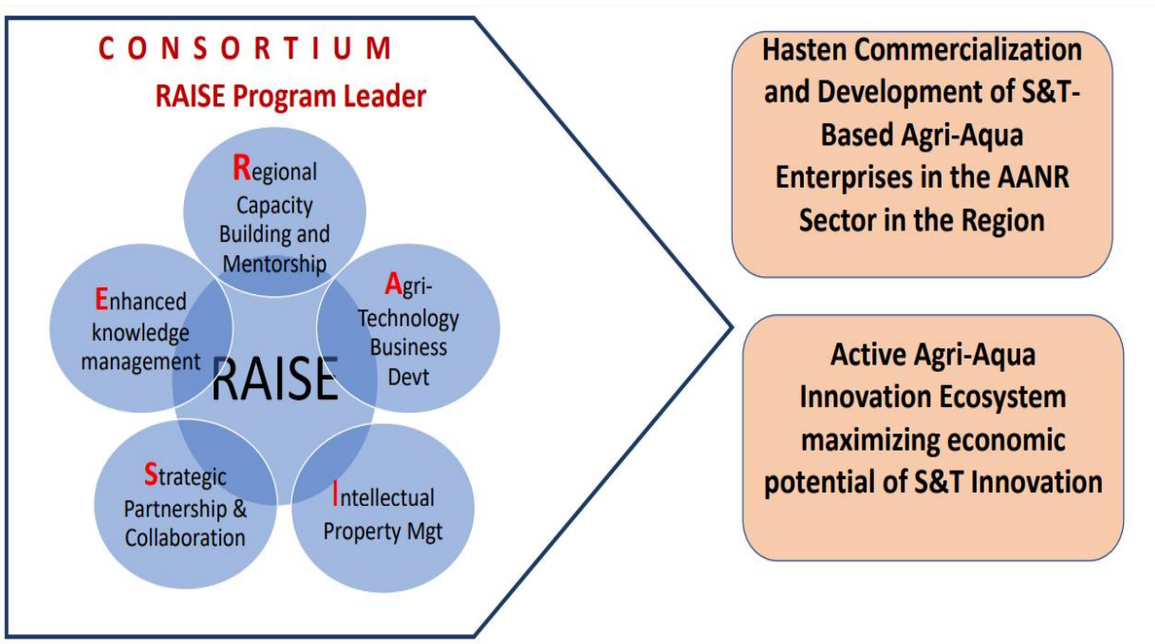


Figure 4. The conceptual framework of RAISE

3. DISCUSSION OF RESULT AND FINDINGS

During the year one implementation of the Intellectual Property and Technology Business Management (IP-TBM) in Partido State University (ParSU) through Regional Agri-Aqua Innovation System Enhancement (RAISE) Program in Bicol, the following are the highlights of accomplishments based on expected outputs:

- ***Under the Publication***, two (2) IEC materials have developed, which are the following: 1) training module on Intellectual Property Management which was developed right after the IP Masterclass training wherein most of the references are based on the lectures given by resource persons; and 2) training module on developing decorative items from abalone shells which this IEC material was an output of a faculty researchers;

- ***Under the Patent and IP filings***, Partido State University IP Unit through the IP-TBM project was able to file eight (8) IP application in agri-aqua technologies through an online IP filing in the Intellectual Property Office of the Philippines (IPOPHL) website. The target output was successfully accomplished despite delay in the fund download;

- ***Under the Products***, ParSU IP Unit through the IP-TBM project was able to complete the following: 1) five (5) prior arts search (PAS) reports prepared for the research proposals of faculty from year 2019 to 2022; 2) conducted the following inventory for IP and commercialization: a) potential IPs of students' and faculty's completed researches at ParSU Tinambac Campus, b) IP assets based from the University's completed researches from year 2014-2022, c) matured technologies submitted to Department of Science and Technology (DOST) V for preliminary formality examination of Fairness Opinion Board (FOB) Secretariat; and d) information of knowledge resources was already submitted to the RAISE Bicol Information Hub; 3) two (2) communication plan was crafted with the following title: a) Local Dried Fish Production in Partido Area, and b) Molding Machine for Ibos (Suman) Wrapper"; and 4) two (2) technologies qualified for technology pitching at the Regional Agri-Aqua Technology Pitching Competition on November 29, 2022 (Coconut Leaf Molding Machine and 3-in-1 Rice Cake making Machine).

- ***Under the Peoples and Services***, ParSU through the IP-TBM project was able to complete the following: 1) send 2 participants for each of the following trainings on IP and commercialization as part of the capacity building activity of the RAISE program: a) Prior Art Search and IP Audit Workshop; b) IP MasterClass; c) Agribusiness MasterClass; d) Technology Commercialization Mentorship Series; d) Technology Promotion Mentorship Series; and e) Communication Plan Workshop; 2) Conducted two (2) re-echo seminars to capacitate the technology transfer personnel of the University which 81 faculty have been trained for IP awareness, patent searching, and patent drafting.

- ***Under the Places and Partnership***, ParSU signed a commitment letter with DOST-PCAARD under the RAISE IP-TBM project; and

- ***Under the Policy***, a comprehensive IP policy for ParSU was developed and approved by ParSU community and the Board of Regents (BOR). With this accomplishment, the IP-TBM personnel will be able to manage the project effectively in accordance with the existing laws on IP.

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4. **Outputs (6Ps)**

During the Year 1 implementation of RAISE IP-TBM in ParSU, the following are the major accomplishment delivered base on the expected outputs: Table 1 shows the list of accomplishment under Publication; table 2 shows the list of Intellectual Property filed under Patents; table 3 shows the list of Prior Arts Search (PAS) report conducted under the Product; table 4 shows the list of trainings attended by CMI staff under People Services;

PUBLICATION:

Table 1. List of Accomplishment under Publication

Title of Module/IEC Developed	Name of Author	Status
Module on Intellectual Property Management	Michael C. Barbecho	Completed
IEC Material for Abalone Shell product	Rosemarie A. Tabardillo	Completed

PATENTS/IP:

Table 2. List of Intellectual Property (IP) filed

Title of IP Filed	Inventor(s)/Maker(s) / Designer(s)	Date of Filing	Category/ Application No.	Status
Solar/Grid Powered Fish Dryer	Michael C. Barbecho Luven A. Haja Jose S. Fernandez Myrna C. Bigueja Irene P. Daet Glenda S. Sales	August 16,2022	Industrial Design 3-2022-050799	Submitted to the IPOPHL the response on the findings in the Formality Examination Report on November 14, 2022
		August 19,2022	Invention 1-2022-050436	Submitted to the IPOPHL the payment for Publication Fee on October 26, 2022 as requirement in the acknowledgement and accordance of filing date
Fish Vendor Cooler	Jose S. Fernandez Leoben C. Formalejo Alvin O. Primavera Gemmah T. Barcillano	August 19,2022	Utility Model 2-2022-050727	Received the Formality Examination Report on September 28, 2022
		August 19,2022	Industrial Design 3-2022-050797	Submitted to the IPOPHL the response on the findings in the Formality Examination Report on November 14, 2022
Abalone Shell Christmas Decoration	Rosemarie A. Tabardillo	August 19,2022	Industrial Design 3-2022-050798	Submitted to the IPOPHL the response on the findings in the Formality Examination Report on November 14, 2022
Abalone Shell Candle Holder	Rosemarie A. Tabardillo Angeli Rose Rioja	September 27 2022	Industrial Design 3-2022-050993	Received the Formality Examination Report on November 14, 2022
Abalone Shell Table Lamp	Rosemarie A. Tabardillo	September 27 2022	Industrial Design 3-2022-050994	
Molding Machine for Coconut Leaf (Ibos) Suman Wrapper	Jose S. Fernandez Leoben C. Formalejo Alvin O. Primavera Gemmah T. Barcillano	September 27 2022	Industrial Design 3-2022-050992	

PRODUCT:

Table 3. List of Prior Arts Search Report Conducted
(2019-2022 Faculty Research Proposals)

Title of Research Proposal	Title of Patentable IP	Researchers	Recommended IP Protection
Design and Development of Smart Solar/Grid Powered Fish Dryer	Solar/Grid Powered Fish Dryer	Michael C. Barbecho Luven A. Haja Jose S. Fernandez Myrna C. Bigueja Irene P. Daet Glenda S. Sales	Invention
Designing and Development of Fish Vendor Cooler	Fish Vendor Cooler	Jose S. Fernandez Leoben C. Formalejo Alvin O. Primavera Gemmah T. Barcillano	Utility Model
Development and Fabrication of 3 in 1 Knapsack Agricultural Sprayer	Solar Powered Knapsack Agricultural Sprayer	Jose S. Fernandez Leoben C. Formalejo	Utility Model
In-vitro Mass Propagation in Different Carbon Sources of Taro	Mass Propagation Method of Taro by In-Vitro Culture	Karen Eyre Moshie O. Artiaga	Utility Model
Metal Foundry Furnace	Development of Metal Foundry Furnace	Engr. Raymundo V. Romero	Utility Model

PEOPLE SERVICES:

Table 4. List of Trainings Attended by CMI Staff

Title of Training	Name of Participant	Start Date	Completion Date
Intellectual Property (IP) Masterclass Webinar	Michael C. Barbecho Jose C. Tunay Jr. III	March 23, 2022	May 27,2022
Agribusiness Masterclass Mentorship Series Webinar	Jenny P. Lorio Gina M. Villareal	October 2022	November 2022
Technology Commercialization Mentorship Series Webinar	Joel P. Sadol Maria France Q. De Vergara	July 2022	September 2022
Technology Promotion Mentorship Series Webinar	Kier P. Gasga Joel P. Sadol	September 2022	November 2022
PAHATIRAN: Communication Action Planning	Michael C. Barbecho	June 14, 2022	June 15, 2022
Regional Knowledge Inventory and Communication Planning Workshop	Michael C. Barbceho Melody G. Perias Jella Marie N. Oyales	October 24, 2022	October 25, 2022

PLACES AND PARTNERSHIP

Partido State University engaged the signing of one (1) commitment letter under the RAISE IP-TBM project. Said commitment letter was signed by the agency head Dr. Raul G. Bradecina and other concerned persons (please see proof of signing in the appendices).

POLICY

The IP Unit of ParSU crafted and developed the PSU IP Policy and was approved by the BOR with Resolution No. 46 s. 2022 during the 85th Regular Board of Regents Meeting July 21, 2022.

5. Outcomes

After the completion of this project, the following are the expected potential outcomes:

- Increased number of IP applications and registered IP for the institution's agri-aqua technologies;
- Integrated DOST-PCAARRD Intellectual Property and Technology Business Management (IP-TBM) functions in the Intellectual Property Unit of the university;
- Trained staff that will be in charge of the IP-TBM-based functions; and
- Create partnerships and agreements for commercialization with technology adaptors.

6. Potential Impacts (2Is)

One of the potential economic impact during the implementation of this project is the availability of IP patent drafter in ParSU after the completion of Intellectual (IP) Masterclass. The patent drafter is a regular faculty designated in the IP Unit of ParSU. All IP patent documents and applications are being prepared in the IP office before filing online. With this, the university does not need an item for a Certified Patent Agent thereby saving money for the salary of personnel in that item.

7. Literature Cited

Basic Invention Disclosure
https://www.wellspring.com/blog/the-basics-of-invention-disclosures#:~:text=At%20its%20core%2C%20an%20invention,it%20is%20created%20and%20reproduced.

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https://globaledge.msu.edu

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https://www.globalinnovationindex.org/

Intellectual Property Rights

https://allea.org/intellectual-property-rights/

IPOPhil Industrial Design Search
https://onlineservices.ipophil.gov.ph/wopublish-search/public/designs?2

IPOPhil Patent and Utility Model Search
https://onlineservices.ipophil.gov.ph/wopublish-search/public/patents?0

J. Serapion 2022. Claim Drafting Lecture. 2022 DOST-PCAARD Intellectual Property Master Class.

J.W. Salvacion. Patent Search Lecture. 2022 DOST-PCAARD Intellectual Property Master Class.

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Technology and IP law Glosarry
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The Revised Implementing Rules and Regulations For Patents, Utility Models And Industrial Designs
https://drive.google.com/file/d/1WFiTWpNmwf-LYW7k8jeNElg4jBH-68uL/view

Types of Intellectual Property Protected by Philippine Law
https://businessdiary.com.ph/18383/types-of-intellectual-properties-protected-by-philippine-law/

8. Problems

During the 1st and 2nd quarter of the implementation of the project, several problems were encountered. Some of the major problems and concerns encountered are: Time constraints of the personnel involved in the project specifically the project leader/project staff level 3 was academically loaded during the 1st and 2nd quarter of the implementation of the project, affecting the close monitoring of the project; the Project Leader/Project Staff Level 3 was overloaded with work due to the absence of Science Research Assistant (SRA) resulting to multitasking during the implementation of the project**;** some of the activities planned such as re-echo trainings, IP filings, and purchase of supplies and equipment were not carried-out on specific schedule due to delayed fund download; and COVID-19 restrictions prevents the personnel involved in this project from reporting for onsite work to access better internet connection, as a result some of the important meetings and updates regarding the project were missed out.

For the solution of the above stated problems and concerns, it is suggested that the project leader/staff level 3 should submit written request for academic de-loading of personnel involve this 3rd and 4th quarter of RAISE IP-TBM project implementation; request the help and assistance of RAISE IP-TBM Program leader Engr. Christoper Pacardo in hiring qualified Science Research Assistant **(**SRA); carry out some of the expected outputs and deliverables that does not require funding such as preparation of IP application documents, Prior art searching, IP inventories, IEC material development, etc., while waiting for the fund to be downloaded; and ook for sites with better internet signal to attend during important meetings and updates regarding the project.

Appendices

(Please see file folder Proof of 6Ps)